

Question 1 (15 marks) Use a separate page/booklet

Marks

(a) Find: $\int x\sqrt{3x-1} dx$ 3

(b) By using the substitution $t = \tan \frac{\theta}{2}$, evaluate

$\int_0^{\frac{\pi}{2}} \frac{d\theta}{2+\sin \theta}$ 3

(c) (i) Split into partial fractions: $\frac{8}{(x+2)(x^2+4)}$ 2

(ii) Hence evaluate: $\int_0^2 \frac{8 dx}{(x+2)(x^2+4)}$ 3

(d) If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x dx$, ($n \geq 2$)

(i) Show that $I_n = (n-1)I_{n-2} - (n-1)I_n$ 2

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x dx$ 2

Question 2 (15 marks) Use a separate page/booklet

Marks

(a) If $z = 3 + 2i$, plot on an Argand diagram

(i) z and $\bar{z}$ 1

(ii) iz 1

(iii) $z(1+i)$ 1

(b) (i) Find all pairs of integers a and b such that $(a+ib)^2 = 8+6i$ 1

(ii) Hence solve: $z^2 + 2z(1+2i) - (11+2i) = 0$ 2

(c) (i) If $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$, find z^6 2

(ii) Plot on an argand diagram, all complex numbers that are the solutions of $z^6 = 1$ 2

(d) Sketch the locus of the following. Draw separate diagrams.

(i) $\arg(z-1-2i) = \frac{\pi}{4}$ 1

(ii) $z\bar{z} - 3(z+\bar{z}) \leq 0$ 2

(iii) $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{3}$ 2

Question 3 (15 marks) Use a separate page/booklet

Marks

- (a) For the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$
- (i) Find the eccentricity. 1
 - (ii) Find the coordinates of the foci S and S'. 1
 - (iii) Find the equations of the directrices. 1
 - (iv) Sketch the curve $\frac{x^2}{25} + \frac{y^2}{16} = 1$ 1
 - (v) Show that the coordinates of any point P can be represented by $(5\cos \theta, 4\sin \theta)$ 2
 - (vi) Show that $PS + PS'$ is independent of the position of P on the curve. 3
 - (vii) Show that the equation of the normal at the point P on the ellipse is $5x \sin \theta - 4y \cos \theta - 9 \sin \theta \cos \theta = 0$ 3
 - (viii) If the normal meets the major axis at L and the minor axis at M, prove that $\frac{PL}{PM} = \frac{16}{25}$ 3

Question 4 (15 marks) Use a separate page/booklet

Marks

- (a) The depth of water in a harbour on a particular day is 8.2 m at low tide and 14.6 m at high tide. Low tide is at $1:05 \text{ pm}$ and high tide is at $7:20 \text{ pm}$.
- The captain of a ship drawing 13.3 m water wants to leave the harbour on that afternoon. Find between what times he can leave. (Assume that the tide changes in SHM.) 5
- (b) If $a > 0$, $b > 0$ and $c > 0$, show that
- (i) $a^2 + b^2 + c^2 - ab - bc - ca \geq 0$ 2
 - (ii) $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$ 2
 - (iii) $(a+b+d)(b+c+d)(c+a+d)(a+b+c) \geq 81abcd$ 2
- (c) Using mathematical induction prove that $(1+x)^n - nx - 1$ is divisible by x^2 for $n \geq 2$, n integer. 4

Question 5 (15 marks) Use a separate page/booklet

Marks

- (a) A concrete beam of length $15m$ has plane sides. Cross-sections parallel to the ends are rectangular. The beam measures $4m$ by $3m$ at one end and $8m$ by $6m$ at the other end.

(i) Find an expression for the area of a cross-section at a distance x metres from the smaller end.

3

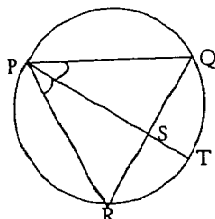
(ii) Find the volume of the beam.

2

- (b) Find the volume of the solid generated by rotating the area bounded by the curve $y = \log_e x$, the x -axis and the line $x = 4$. Use the method of *cylindrical shells*. Rotate the area about the y -axis and give your answer correct to 1 decimal place.

4

(c)



In the diagram, the bisector of the angle RPQ meets RQ in S and the circum-circle of the triangle PQR in T .

(i) Prove that the triangles PSQ and PRT are similar.

2

(ii) Show that $PQ \times PR = PS \times PT$

2

(iii) Prove that $PS^2 = PQ \times PR - RS \times SQ$

2

Question 6 (15 marks) Use a separate page/booklet

Marks

- (a) A point is moving in a circular path about O .

(i) Define the angular velocity of the point with respect to O , at any time t .

1

(ii) Derive expressions for the tangential and normal accelerations of the point at any time t .

4

- (b) A light inextensible string OP is fixed at the end O and is attached at the other end P to a particle of mass m which is moving uniformly in a horizontal circle whose centre is vertically below and distant x from O .

(i) Prove that the period of this motion is $2\pi\sqrt{\frac{x}{g}}$, where g is the acceleration due to gravity.

3

(ii) If the number of revolutions per second is increased from 2 to 3, find the change in x . (Take $g = 10 \text{ m/s}^2$)
Give your answer correct to the nearest millimetre.

3

- (c) The tangent at $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets a directrix at Q . S is the corresponding focus.

Given that the equation of the tangent at P is $bx - ay \sin \theta = ab \cos \theta$:

(i) Find the coordinates of Q .

2

(ii) Show that PQ subtends a right angle at S .

2

Question 7 (15 marks) Use a separate page/booklet

Marks

- (a) Given $y = \frac{x^3}{x^2 - 4}$
- (i) Find the coordinates of all stationary points. 2
- (ii) Find the points of intersection with the coordinate axes and the position of all asymptotes. 2
- (iii) Hence sketch the curve $y = \frac{x^3}{x^2 - 4}$ 2
- (b) Use the graph $y = \frac{x^3}{x^2 - 4}$ to find the number of roots of the equation $x^3 - k(x^2 - 4) = 0$ for varying value of k . 2
- (c) Sketch the following curves:
- (i) $y = \log_e (x + 1)$ 2
- (ii) $y = \log_e \{x + 1\}$ 1
- (iii) $y = |\log_e (x + 1)|$ 1
- (iv) $y = \frac{1}{\log_e (x + 1)}$ 3

Question 8 (15 marks) Use a separate page/booklet

Marks

- (a) Find a polynomial $p(x)$ with real coefficients having $3i$ and $1 + 2i$ as zeros. 3
- (b) A body is projected vertically upwards from the surface of the Earth with initial speed u . The acceleration due to gravity at any point on its path is inversely proportional to the square of its distance from the centre of the Earth.
- (i) Prove that the speed v at any position x is given by $v^2 = u^2 + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right)$ 3
- (ii) Prove that the greatest height H above the Earth's surface is given by $H = \frac{u^2 R}{2gR - u^2}$ 3
- (iii) Show that the body will escape from the Earth if $u \geq \sqrt{2gR}$ 1
- (iv) Find the minimum speed in km/s with which the body must be initially projected from the surface of the Earth so as to never return. (Take $R = 6400 km$, $g = 10 m/s^2$) 1
- (v) If $u = \sqrt{2gR}$, prove that the time taken to reach a height $3R$ above the surface of the Earth is $\frac{14}{3} \sqrt{\frac{R}{2g}}$. 4

(a) Let $u = 3x-1$, $du = 3dx$

$$I = \int \frac{u+1}{3} \sqrt{u} \cdot \frac{1}{3} du$$

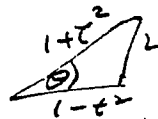
$$= \frac{1}{9} \int u^{3/2} + u^{1/2} du$$

$$= \frac{1}{9} \left(\frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} \right) + C$$

$$= \frac{2}{9} \left(\frac{(3x-1)^{5/2}}{5} + \frac{(3x-1)^{3/2}}{3} \right) + C \quad (3)$$

b) Let $t = \tan \frac{\theta}{2}$

$$\therefore dt = \frac{1+t^2}{2} d\theta$$



$$\tan \theta = \frac{2t}{1-t^2}$$

$$I = \int_0^1 \frac{1}{2+t^2} \times \frac{2dt}{1+t^2}$$

$$= \int_0^1 \frac{dt}{t^2+t+\frac{1}{4}}$$

$$= \int \frac{dt}{\left(t+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left(\tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right)$$

$$= \frac{2}{\sqrt{3}} \left(\frac{\pi}{3} - \frac{\pi}{6} \right)$$

$$= \frac{\sqrt{3}\pi}{9} \quad (3)$$

(i) $\frac{8}{(x+2)(x^2+4)} = \frac{a}{x+2} + \frac{bx+c}{x^2+4}$

res up $\Rightarrow a = 1$

$$8 = x^2+4 + bx^2+2bx+cx+2c$$

$b = -1, c = 2$

$$xp = \frac{1}{x+2} + \frac{-x+2}{x^2+4}$$

$$= \frac{1}{x+2} - \frac{1}{2} \frac{2x}{x^2+4} + \frac{2}{x^2+4} \quad (2)$$

$$I = \int \frac{1}{x+2} - \frac{1}{2} \frac{2x}{x^2+4} + \frac{2}{x^2+4} dx$$

$$= \left[\ln|x+2| - \frac{1}{2} \ln|x^2+4| + 2 \cdot \frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2$$

$$= \ln 4 - \frac{1}{2} \ln 8 + \frac{\pi}{4} - \ln 2 + \frac{1}{2} \ln 4$$

$$= \ln \sqrt{2} + \frac{\pi}{4} \quad (3)$$

$$I_n = \int_0^{\pi/2} \cos^{n-1} x \cdot d(\sin x)$$

$$= \left[\sin x \cdot \cos^{n-1} x \right]_0^{\pi/2} - \int_0^{\pi/2} \sin x \cdot (n-1) \cos^{n-2} x \cdot \sin x dx$$

$$= 0 + \int_0^{\pi/2} (n-1) \cos^{n-2} x \cdot (1 - \cos^2 x) dx$$

$$\therefore I_n = (n-1) I_{n-2} \quad (2)$$

$$\therefore I_n = \frac{n-1}{n} I_{n-2}$$

$$\therefore I_6 = \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot I_0$$

$$I_0 = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$$

$$\therefore I_6 = \frac{5\pi}{16} \quad (2)$$



(b) $a^2 - b^2 = 8$ & $ab = 3$ (1)

$$\therefore a = 3, b = 1 \text{ or } a = -3, b = -1$$

$$z = \frac{-2(1+2i) \pm \sqrt{4(1+2i)^2 + 4(11+2i)}}{2}$$

$$= -1-2i \pm \sqrt{8+6i}$$

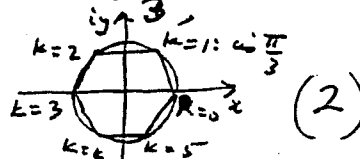
$$= -1-2i+3+i \text{ or } -1-2i-3-i$$

$$= 2-i \text{ or } -4-3i \quad (2)$$

C(i) $z^6 = 1$ by de Moivre's
 $= 1$ (2) Reason

(ii) $z = \cos \frac{2\pi k}{6}$

$$= \cos \frac{\pi k}{3} \quad k=0,1,2,3,4,5$$



(d) (i) $z = x+iy$ (1)

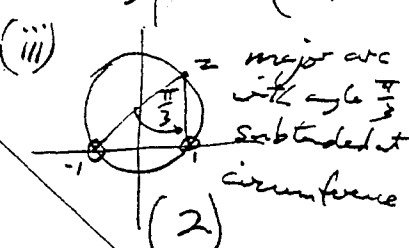
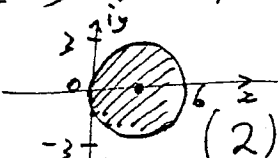


(ii) Let $z = x+iy$

$$\therefore x^2+y^2-3(2x) \leq 0$$

$$\therefore x^2-6x+9+y^2 \leq 0+9$$

$$\therefore (x-3)^2+y^2 \leq 9$$



3(a) (i) $a=5, b=4$

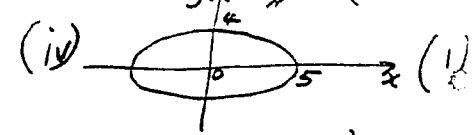
$$b^2 = a^2(1-e^2)$$

$$\therefore e^2 = \frac{9}{25}$$

$$\therefore e = \frac{3}{5} \text{ or } 0.6 \quad (1)$$

(ii) $S(3,0) S'(-3,0)$ (1)

(iii) $x = \pm \frac{25}{3}$ (1)



(v) $S, b, S, LHS = \frac{25 \cos^2 \theta}{25} + \frac{16 \sin^2 \theta}{16}$
 $= 1 = RHS$ (2)

(vi) $PS + PS' = ePM + ePM'$

$$= e \left(\frac{25}{3} - 5 \cos \theta \right) + e \left(5 \cos \theta \right)$$

$$= \frac{25}{3} \times \frac{50}{3} \quad (3)$$

$$= 10, \text{ independent}$$

(vii) Now $x = 5 \cos \theta + y = 4 \sin \theta$

$$\therefore \frac{dx}{d\theta} = -5 \sin \theta, \frac{dy}{d\theta} = 4 \cos \theta$$

$$\therefore \frac{dy}{dx} = -\frac{4 \cos \theta}{5 \sin \theta}$$

$$\therefore \text{normal: } \frac{y-4 \sin \theta}{x-5 \cos \theta} = \frac{5 \sin \theta}{4 \cos \theta} \quad (3)$$

$$\therefore 5x \sin \theta - 4y \cos \theta - 9 \sin \theta \cos \theta = 0$$

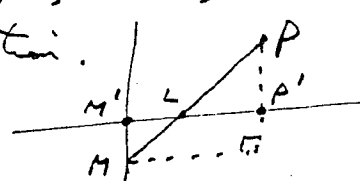
(viii) Let $y > 0, x = \frac{9 \cos \theta}{5}$

$$\therefore L \left(\frac{9 \cos \theta}{5}, 0 \right)$$

Let $x=0, y = -\frac{9}{4} \sin \theta \therefore M \left(0, -\frac{9}{4} \sin \theta \right)$

$$\therefore \frac{PL}{PM} = \frac{5 \cos \theta - \frac{9 \cos \theta}{5}}{5 \cos \theta} \text{ using}$$

x -values only. [This all that is necessary because, for similar triangles, the sides are in proportion.]

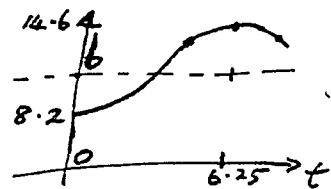


$$\therefore \frac{PL}{PM} = \frac{P'L}{P'M}$$

$$\therefore \frac{PL}{PM} = \frac{25-9}{25} = \frac{16}{25} \quad (3)$$

4(a) Now, as in SHM,

$$x = -a \cos \omega t + b \text{ and } T = \frac{2\pi}{\omega}$$



$$b = \frac{14.6 + 8.2}{2} = 11.4$$

$$\therefore x = -a \cos \omega t + 11.4$$

Time from 1:05 to 7:20 is 6.25h

$$\therefore \omega = \frac{2\pi}{12.5}, \text{ using } \omega = \frac{2\pi}{T}$$

$$\therefore x = -a \cos \frac{4\pi}{25} t + 11.4$$

$$a = \frac{14.6 - 8.2}{2} = 3.2$$

$$\therefore x = -3.2 \cos \frac{4\pi}{25} t + 11.4$$

$$\text{at } x = 13.3, \cos \frac{4\pi}{25} t = -0.59 = -\frac{19}{32}$$

$$\therefore t = 4.3897 = 4\text{h } 23\text{min}$$

$\therefore$ first time is 5:28pm

$$\text{and time, } t = 6.25 + (6.25 - 4.39) = 8.11 = 8\text{h } 6.6\text{min}$$

2nd time is 9:12pm (5)

$$1) (i) a^2 + b^2 \geq 2ab \therefore (a+b)^2 \geq 0$$

$$\& c^2 + b^2 \geq 2bc$$

$$2) a^2 + c^2 \geq 2ac$$

$$\therefore 2(a^2 + b^2 + c^2) \geq 2(ab + bc + ac)$$

$$a^2 + b^2 + c^2 - ab - bc - ac \geq 0 \quad (2)$$

$$i) \text{ Now } (a^2 + b^2 + c^2 - ab - bc - ac) \geq 0$$

$$a^3 + b^3 + c^3 - 3abc \geq 0$$

$$\therefore \frac{a^3 + b^3 + c^3}{3} \geq abc$$

$$\text{Let } a = A^{1/3}, b = B^{1/3}, c = C^{1/3}$$

$$\frac{A + B + C}{3} \geq A^{1/3} B^{1/3} C^{1/3}$$

$$\frac{A + B + C}{3} \geq \sqrt[3]{ABC}$$

$$a + b + c \geq \sqrt[3]{abc} \quad (2)$$

4(b)(iii)

$$\begin{aligned} LHS &= (a+b+d)(b+c+d)(c+a+d)(a+b+c) \\ &\geq 3^3 \sqrt[3]{abd} \cdot 3^3 \sqrt[3]{bcd} \cdot 3^3 \sqrt[3]{cad} \cdot 3^3 \sqrt[3]{abc} \\ &\geq 81 \cdot \sqrt[3]{a^3 b^3 c^3 d^3} \\ &\geq 81 \cdot abcd = RHS \quad (2) \end{aligned}$$

(c) Step 1 Let $n=2, (1+x)^2 - 2x - 1 = x^2$ which is divisible by x^2 .

Step 2 Suppose $(1+x)^k - kx - 1 = Mx^2$ where $M = M(x)$, a poly in x

$$\therefore (1+x)^k = Mx^2 + kx + 1$$

Step 3 RTP $(1+x)^{k+1} - (k+1)x - 1$ is divisible by x^2

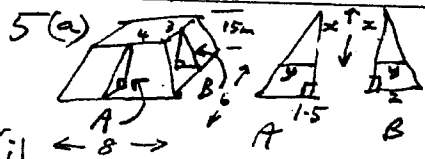
$$\begin{aligned} \text{Now exp} &= (1+x)(1+x)^k - kx - x - 1 \\ &= (1+x)[Mx^2 + kx + 1] - kx - x - 1 \\ &= Mx^2 + kx^2 + Mx + 1 - kx - x - 1 \\ &= Mx^2 + kx^2 + Mx \\ &= x^2(M + k + Mx) \end{aligned}$$

which is divisible by x^2

Step 4: Statement true for $n=1, (4)$

Using Steps 2 & 3, $\therefore$ true for $n=1+1=2$

Similarly, true for $n=3, 4$ and so on



(i) Consider a horizontal cross-section x m from the top. Using diagrams A+B and similar triangles:

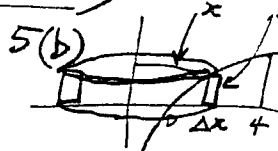
$$\frac{x}{15} = \frac{y_A}{1.5} \& \frac{x}{15} = \frac{y_B}{2}$$

$$\therefore y_A = \frac{x}{10} \& y_B = \frac{2x}{15}$$

$$\therefore \text{cross section} = (3 + 2y_A) \times (4 + 2y_B)$$

$$\begin{aligned} \therefore A &= \left(3 + \frac{x}{5}\right) \left(4 + \frac{4x}{15}\right) \\ &= 12 \left(1 + \frac{2x}{15} + \frac{x^2}{225}\right) \\ &= 12 + \frac{8x}{5} + \frac{4x^2}{75} \quad (3) \end{aligned}$$

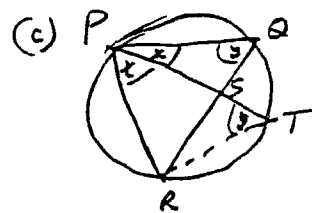
$$\begin{aligned} (ii) V &= \int_0^{15} \left(12 + \frac{8x}{5} + \frac{4x^2}{75}\right) dx \\ &= \left[12x + \frac{8x^2}{10} + \frac{4x^3}{225}\right]_0^{15} \\ &= 420 \text{ m}^3 \quad (2) \end{aligned}$$



$$\Delta V = 2\pi x \cdot \Delta x \cdot L$$

(ignoring 2nd order dy)

$$\begin{aligned} \therefore V &= \lim_{\Delta x \rightarrow 0} \sum 2\pi x \cdot \Delta x \cdot L \\ &= \int_0^4 2\pi x \cdot L \cdot dx \\ &= 2\pi \int_0^4 L \cdot x \cdot d\left(\frac{x^2}{2}\right) \\ &= 2\pi \left[\frac{x^2}{2} \cdot L\right]_0^4 - \pi \int_0^4 x^2 \cdot dL \\ &= 2\pi \cdot 8L_4 - 2\pi \int_0^4 x^2 \cdot dL \\ &= 16\pi L_4 - \pi \left[\frac{x^2}{2} \cdot L\right]_0^4 \\ &= \pi \left(16L_4 - \frac{1}{2} \cdot 16\right) \\ &= 46.1 \text{ m}^3 \quad (4) \end{aligned}$$



(i) $\angle PQR = \angle QSR = y$ (at circum from common $x = \angle RPT = \angle RST$ (90 $\therefore \Delta PQS \parallel \Delta PTR$ (2 corresponding angles equal, x & y))

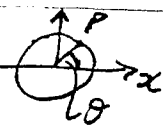
$$(ii) \text{ Using ratios of similar } \frac{PQ}{PT} = \frac{PS}{PR} = \frac{QS}{TR}$$

$$\therefore PQ \times PR = PS \times PT$$

$$(iii) RHS = PQ \times PR - RS \times SQ$$

$$\begin{aligned} \text{Now } PQ \times PR &= PS \times PT \quad (ii) \text{ also} \\ \text{and } RS \times SQ &= PS \times ST \quad (\text{prod of intercepts of 2 intersecting ch}) \\ \therefore RHS &= PS \times PT - PS \times ST \\ &= PS(PT - ST) \\ &= PS \times PS \\ &= PS^2 \end{aligned}$$

$$\therefore LHS = RHS \quad (2)$$

6(a)(i)  $\text{ang. vel. } w = \frac{d\theta}{dt}$

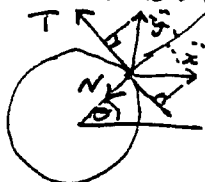
(ii) $x = r \cos \theta$
 $\therefore \dot{x} = -r \sin \theta \cdot w$

$\ddot{x} = -w \cdot r \cos \theta \cdot w - r \sin \theta \cdot \dot{w}$

Let $y = r \sin \theta$

$\therefore \dot{y} = r \cos \theta \cdot w$

$\ddot{y} = w \cdot r \sin \theta \cdot w + r \cos \theta \cdot \dot{w}$



$T = \ddot{y} \cos \theta - \dot{x} \sin \theta$

$= -r w^2 \cos \theta \sin \theta + r w^2 \sin \theta \cos \theta$
 $+ r w^2 \cos \theta \sin \theta + r \sin \theta \dot{w}$
 $= r \dot{w}$

$N = -\ddot{y} \sin \theta - \dot{x} \cos \theta$

$= -r w^2 \sin^2 \theta - r \sin \theta \cos \theta \cdot w$
 $+ r w^2 \cos^2 \theta + r \sin \theta \cos \theta \cdot w$

$= r w^2 \cos 2\theta$ (4)



Horizontally mg Vertically
 $r w^2 = T \sin \theta$ & $0 = T \cos \theta - mg$

$\therefore \frac{r w^2}{g} = \tan \theta$

$\therefore w = \sqrt{\frac{g \tan \theta}{r}}$

At $\tan \theta = \frac{r}{x}$

$\therefore w = \sqrt{\frac{g}{x}}$

Let $T = \frac{2\pi}{T}$ (period)

$= 2\pi \sqrt{\frac{x}{g}}$ (3)

$f = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{g}{x}}$

$2 = \frac{1}{2\pi} \sqrt{\frac{g}{x_1}} \quad 3 = \frac{1}{2\pi} \sqrt{\frac{g}{x_2}}$

$x_1 = \frac{g}{16\pi^2} \quad \& \quad x_2 = \frac{g}{36\pi^2}$

$\approx 0.0633 \quad \& \quad x_2 = 0.02814$

difference $\approx 0.035 \text{ m}$ (3)

(continued)

6(c)(i) Directrix $x = \frac{a}{e}$

Sub into tangent

$b(\frac{a}{e}) - ay \sin \theta = ab \cos \theta$

$\therefore y = \frac{\frac{ab}{e} - ab \cos \theta}{a \sin \theta}$
 $= \frac{b(1 - e \cos \theta)}{e \sin \theta}$

$\therefore Q(\frac{a}{e}, \frac{b(1 - e \cos \theta)}{e \sin \theta})$ (2)

(ii) $m_{PS} = \frac{b \tan \theta - 0}{a \sec \theta - ae}$

$\& \quad m_{QS} = \frac{\frac{b(1 - e \cos \theta)}{e \sin \theta} - 0}{\frac{a}{e} - ae}$

$\therefore m_{PS} m_{QS} = \frac{b \tan \theta}{a \sec \theta - ae} \times \frac{be(1 - e \cos \theta)}{ae \sin \theta (1 - e^2)}$

multiply top & bot. by $\cos \theta$

$\therefore m_{PS} m_{QS} = \frac{b \sin \theta}{a(1 - e \cos \theta)} \times \frac{be(1 - e \cos \theta)}{ae \sin \theta (1 - e^2)}$

Now $b^2 = a^2(e^2 - 1)$

$\therefore m_{PS} m_{QS} = -1$, rt angle at S. (2)

7(a)(i) $\frac{dy}{dx} = \frac{(x^2 - 4) \cdot 3x^2 - x^3 \cdot 2x}{(x^2 - 4)^2}$
 $= \frac{x^4 - 12x^2}{(x^2 - 4)^2}$

$\therefore$ stationary $y' = 0 \Rightarrow (x^2 - 12)x^2 = 0$

$\therefore x = 0$ or $x = \pm 2\sqrt{3}$

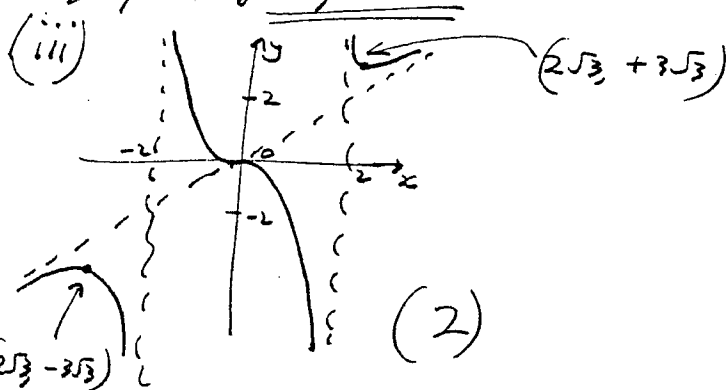
$\therefore$ pts $(0, 0)$ & $(2\sqrt{3}, \pm 3\sqrt{3})$ (2)

(ii) $x^2 - 4 \overline{) x^3 - 4x}$
 $\underline{x^3 - 4x}$
 0

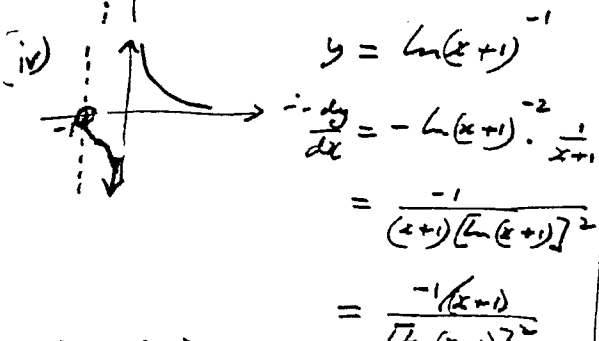
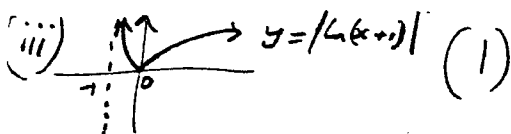
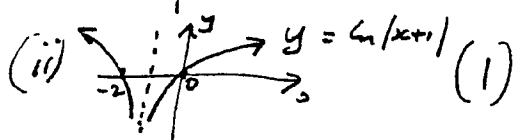
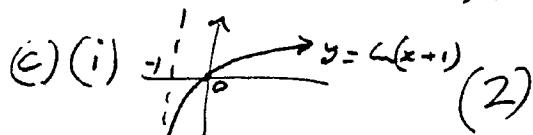
$\therefore y = x + \frac{4x}{(x-2)(x+2)}$

points intersection axes $(0, 0)$ (2)

asymptotes $y = x$, $x = \pm 2$



7(b) Consider the two graphs of
 $y = k$ & $y = \frac{x^3}{x^2 - 4}$ for $k > 3\sqrt{3}$, 3 roots
 $\therefore \frac{x^3}{x^2 - 4} = k$.
 $k = 3\sqrt{3}$, 2 roots
 $-3\sqrt{3} < k < 3\sqrt{3}$, 1 root
 $k = -3\sqrt{3}$, 2 roots
 $k < -3\sqrt{3}$, 3 roots (2)



Using L'Hopital's rule for $x \rightarrow -1$

$$\lim_{x \rightarrow -1} \frac{dy}{dx} = \lim_{x \rightarrow -1} \frac{1/(x+1)^2}{2 \ln(x+1)/x+1}$$

$$= \lim_{x \rightarrow -1} \frac{1/(x+1)}{2 \ln(x+1)}$$

Using L'Hopital's rule again

$$\lim_{x \rightarrow -1} \frac{dy}{dx} = \lim_{x \rightarrow -1} \frac{-1/(x+1)^2}{2/(x+1)}$$

$$= \lim_{x \rightarrow -1} \frac{-1}{2(1+x)}$$

$$= -\infty. \text{ (See graph above)}$$

Graph does not include $(-1, 0)$
 but vertical tangent at $(-1, 0)$ as
 shown by L'Hopital's rule

(3)

1 for general graph

1 for $(0, 0)$ not included

1 for $x = -1$ not included on dy

8(a) If $3i$ is a root so is $-3i$, similarly $1+2i$ & $1-2i$
 Note $(x-a-bi)(x-a+bi) = x^2 - 2\operatorname{Re}(a)x + |a|^2$
 $\therefore (x-3i)(x+3i) = x^2 + 9$ & $(x-1-2i)(x-1+2i) = x^2 - 2x + 5$
 $\therefore P(x) = (x^2 + 9)(x^2 - 2x + 5)$
 $= x^4 - 2x^3 + 14x^2 - 18x + 45$ (3)

(b) (i)
 $\frac{d(\frac{1}{2}u^2)}{dx} = \frac{-k}{x^2}$
 $\therefore u du = -kx^{-2} dx$
 $\therefore \int u du = \int -kx^{-2} dx$
 $\frac{u^2}{2} = \frac{k}{x} + \frac{C}{2}$

$$\therefore \frac{v^2}{2} - \frac{u^2}{2} = \frac{k}{x} - \frac{k}{R}$$

At $x=R$, $u=0 \therefore \frac{-k}{R^2} = \frac{C}{2}$
 $\therefore k = gR^2$

$$\therefore v^2 = \frac{2gR^2}{x} + u^2 - \frac{2gR^2}{R}$$

$$\therefore v^2 = u^2 + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right)$$
 (3)

(ii) At greatest height, $v=0$ and $x=R+H$

$$\therefore 0 = u^2 + 2gR^2 \left(\frac{1}{R+H} - \frac{1}{R} \right)$$

$$\therefore u^2 = 2gR \left(\frac{H}{R+H} \right)$$

$$\therefore H \left(\frac{2gR}{u^2} - 1 \right) = R$$

$$\therefore H = \frac{u^2 R}{2gR - u^2}$$
 (3)

(iii) Let $H \rightarrow \infty$,

$$\therefore u^2 = 2gR$$
 (1)

$$\therefore u \geq \sqrt{2gR} \text{ for escape.}$$

(iv) $u = \sqrt{\frac{2 \times 10 \times 6400}{1000}}$
 $= 11.3 \text{ km/s}$ (1)

(v) Now $u = \sqrt{2gR}$

$$\therefore v^2 = 2gR + 2gR^2 \left(\frac{1}{x} - \frac{1}{R} \right)$$

$$= 2gR \left(1 + \frac{R}{x} - 1 \right)$$

$$= \frac{2gR^2}{x}$$

(v) Cont.
 $\therefore v = \sqrt{\frac{2gR^2}{x}}$

$$\therefore \frac{dt}{dx} = \frac{1}{R\sqrt{g}} \cdot x^{-1/2}$$

$$\therefore \int_0^T dt = \frac{1}{R\sqrt{g}} \int_R^{4R} x^{-1/2} dx$$

$$\therefore T = \frac{1}{R\sqrt{g}} \cdot \frac{2}{3} [x^{3/2}]_R^{4R}$$

$$= \frac{1}{3R\sqrt{g}} (4R)^{3/2} - R$$

$$= \frac{1}{3} \sqrt{\frac{2}{g}} \cdot \sqrt{R} \cdot 7$$

$$= \frac{7}{3} \sqrt{\frac{2R}{g}}$$

$$= \frac{14}{3} \sqrt{\frac{R}{2g}}$$
 (4)